

One idea that is used today is the split keyboard. These range from the gently humped Microsoft Natural Keyboard that only suggests a split, to the extreme Kinesis keyboard where the keys are grouped in two discrete depressions in the keyboard surface, or the [ref.5] Goldtouch keyboard with its adjustable central hinge. The idea with all these keyboards is to place the keys where they are most easily accessible to the fingers of each hand. However, the more extreme designs do require full touch typing skills because the keys are more difficult to see.

When using a normal flat keyboard a wrist rest is essential to place the heels of the palms on the same level as the keying surface. The [ref.6] TrulyErgonomic keyboard, which has yet to go into production, incorporates such a rest.

Other input devices

Various attempts have been and are being made to augment or replace the keyboard. The mouse is now ubiquitous, but like touch-screens and light pens, compared to the keyboard it is a point and select device, rather than a data input device. The influence of ergonomic design is very evident in the evolution of the mouse. The first mouse, [ref.7] made in 1964 by Douglas Engelbart, was a rectangular wooden box with two wheels mounted at right angles underneath. Production versions of this mouse weren't much better.

Interestingly Engelbart points out that there is a difference between designing for ease-of-use and for high performance. He envisaged knowledge workers of the future being highly skilled with interface systems that were not necessarily easy to use. This is perhaps demonstrated by the number of buttons found on different mouse designs. Apple favour the simple single button, while other designs have four or five buttons, plus perhaps a wheel. With more buttons it is often more difficult to remember their functions and to manipulate them correctly, at speed.

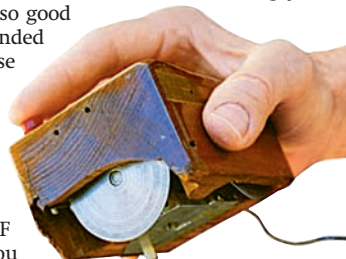
Modern mice look very different to Engelbart's prototype, being ergonomically contoured to fit the hand. Although mice seem to be exclusively designed for right-hand use an internet search should reveal quite a few sources for left-handed mice and keyboards and these are of obvious benefit to those who are strongly sinister (left-handed). There are also good arguments to be made for right-handed people sometimes using their mouse over on the left-hand side of the keyboard, thereby balancing the strain on the arms and hands. It takes a little getting used to, but probably carries benefits. It is also worth pointing out that the home keys on the keyboard, F and J, should be right in front of you as you type - dead centre. However, they are significantly to the left of centre of modern keyboards, which are therefore most comfortably used with the mouse on the left. This is a case of not well-considered design, given that most people are right-handed.



Goldtouch split hinged keyboard



TrulyErgonomic keyboard



Original prototype mouse built by Douglas Engelbart

Education and exercise

Ultimately, individual workers, once made aware of a few simple ergonomic principles, are best placed to know whether or not their working conditions are comfortable. Often there are simple actions they can take to improve their work space by, for example, re-locating their display or shading local light sources.

Computer use encourages sitting in one place for long periods of time. It's highly advisable to get into the habit of taking breaks throughout the work day to walk around, or do some stretching exercises.

Furniture

Chairs have acquired high status in the design world, international architect Mies Van der Rohr is almost as famous for his 'Barcelona' chair design as he is for his buildings. Perhaps as a consequence there's more effort expended on designing 'ergonomic' chairs, than any other item of office furniture. One of the most well known designs is the [ref.8] Aeron chair designed in 1994 by Don Chadwick and Bill Stumpf and made by Herman Miller, the American furniture manufacturer that is closely associated with many famous chair designs.

The Aeron comes in three sizes and is fully adjustable in height, tilt and armrest height. It has a lumbar support and the back rest and seat are made from a tough mesh, allowing air to circulate around the user's body. The design flexibility of the Aeron chair comes at a high price. Fortunately, there are many less expensive chairs that can be adjusted for proper user comfort.

A good chair, adjusted for correct posture and support, is essential. Proper lumbar (lower back) support for the natural in-curve of the spine is important. When seated relaxed and upright, with the hands in typing position, the arms and legs should form right angles at upper and forearm and at thigh and calf. The feet should be resting flat on the floor (a foot rest can be used if they aren't). Shoulders should be relaxed and not hunched. The mouse should be easily grasped without much arm extension and the elbow and wrist well supported by the chair arm rest and desk.

For computer work, a wrap around desk with a separate adjustable keyboard support is the best choice. Ideally the keyboard needs to be quite low down, almost in the users lap; placing the keyboard on top of a conventional desk often means it's a little too high.

In conclusion

To sum up — an ergonomically designed workstation environment should be comfortable. If you find you are developing any aches and pains after working at your computer for any length of time something is wrong with the way you have your work space arranged. ■

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A typical modern mouse, the Microsoft Wireless Laser Mouse 5000



The Aeron chair from Herman Miller

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